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A Network Analysis of Empathy: Examining Gender Differences Using the Interpersonal Reactivity Index

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Abstract

The Interpersonal Reactivity Index (IRI) is one of the most widely used measures of empathy, yet its factor structure has shown inconsistent fit across studies. Network analysis offers a complementary, item-level perspective. This study aimed to replicate the empathy network reported by Briganti et al. (2018) in a new sample and to test whether the network structure of empathy differs between women and men. Using an openly available dataset (Han, 2024), we analysed the 28 IRI items from 1,468 university students in the United States (85.49% women; mean age = 21.92 years). A regularised partial-correlation network (EBICglasso) was estimated, and the women's and men's networks were compared with the Network Comparison Test and equal-n subsampling. The full-sample network broadly replicated the expected community structure mirroring the IRI subscales, although a fifth community was detected, with affective items most central. Descriptively, women showed higher centrality and global strength than men, but these differences were not significant, and the men's network was poorly estimated. An omnibus NCT indicated that the overall network structures differed significantly, but the predicted gender differences in centrality and global strength were not supported after correction. Overall, the empathy network was largely replicated. Gender differences in empathy may reflect mean-level rather than structural differences, though sample imbalance warrants caution.

Introduction

Empathy as a Multidimensional Construct

Empathy is widely considered a fundamental capacity for social functioning: it enables a person to register and partly share what someone else is feeling or thinking while still recognising that the experience belongs to the other and not to the self (Decety & Jackson, 2004; Zaki & Ochsner, 2012). Decety and Jackson (2004) capture this in a definition that has become a reference point, treating empathy as the conjunction of three components: an affective sharing of another's state, an understanding of that state, and a preserved distinction between self and other. This definition implies that empathy is not a single ability. It is better understood as a set of related but separable processes that interact dynamically as a situation unfolds (Davis, 1980, 1983; Decety & Jackson, 2004; Raimondi et al., 2023). Two of these components are well established in the literature. Affective empathy refers to vicariously feeling what another person feels, while cognitive empathy refers to understanding another's position and adopting their point of view (Decety & Jackson, 2004; Zaki & Ochsner, 2012). Some accounts add a motivational or prosocial element, a concern or sympathy oriented towards the other's welfare, alongside the awareness and emotion-regulation processes that Decety and Jackson (2004) build into their model (Zaki & Ochsner, 2012). Critically, these components do not operate independently of one another. Earlier work tended to treat the affective and cognitive sides as non-overlapping systems, but later evidence shows them coactivating and becoming functionally coupled once people are in genuine social encounters, each one shaping and modulating the other (Christov-Moore et al., 2014; Decety & Jackson, 2004; Zaki & Ochsner, 2012).

The Interpersonal Reactivity Index

The Interpersonal Reactivity Index (IRI) is among the most widely used self-report measures of empathy. It was designed deliberately as a multidimensional instrument, on the premise that collapsing everything into a single empathy score would hide meaningful differences between its parts (Cliffordson, 2002; Davis, 1980, 1983). The IRI is built from four subscales. Perspective Taking (PT) measures the tendency to spontaneously adopt other people's points of view and is regarded as a cognitive facet. Fantasy (FS) concerns the tendency to identify with fictional characters in books and films; it is conventionally grouped as a cognitive facet, though it relates more closely to emotional responding than that label implies, which is why Paulus (2009) instead treats it as affective. Empathic Concern (EC) captures other-oriented warmth and compassion for those in difficulty, and Personal Distress (PD) captures the self-oriented anxiety and unease people feel when confronted with another's suffering (Davis, 1980, 1983). These four subscales are commonly combined into a cognitive pair (PT, FS) and an affective pair (EC, PD). This split is a useful simplification, but this two-factor reduction fits the data poorly (Raimondi et al., 2023).

The instability is evident in the measure's psychometric record. For all its popularity, the IRI has never settled on a stable factor structure: one-, two-, three-, four-, and five-factor solutions, along with hierarchical models, have all been proposed (Paulus, 2009; Raimondi et al., 2023). A recent meta-analytic structural equation modelling (MASEM) study pooled eleven studies and almost ten thousand respondents (Raimondi et al., 2023). Across these studies the original four-factor model showed inconsistent fit, and collapsing the scales into a clean cognitive-versus-affective pair was no better. The four subscales stayed distinct rather than reduced to two. Part of what drives this is exactly the interdependence described earlier. Because affective and cognitive items are entangled at the item level, the IRI produces sizable cross-loadings and does not fit the simple structure that confirmatory factor analysis (CFA) demands (Grevenstein, 2020; Lucas-Molina et al., 2017). That is why exploratory structural equation modelling (ESEM), which allows those cross-loadings, has repeatedly outperformed CFA on this instrument (Grevenstein, 2020; Lucas-Molina et al., 2017). Findings like these have pushed researchers to look beyond models built on hidden common factors, towards methods that model the connections among the items directly (Borsboom & Cramer, 2013; Briganti et al., 2018).

A Network Approach to Empathy

One way to address the IRI's persistent factor-analytic difficulties is to abandon the assumption that items are mere reflections of an underlying trait. The latent-variable paradigm treats each questionnaire item as an interchangeable indicator whose response is caused by a common factor; the variance in that factor produces the covariation among items, and the items themselves are theoretically passive (Borsboom & Cramer, 2013). A total or subscale score then stands in for the latent cause, collapsing item-level information into a single number. The network paradigm inverts this logic. Rather than positing a hidden common cause, it treats the items as active elements that covary because they directly influence one another, the focus shifts from a hidden common cause to the pattern of direct relationships among items (Borsboom & Cramer, 2013). This is not a claim that latent causes do not exist; it is a shift towards the model likely to be more informative when items interact in complex, asymmetric ways.

A network model rests on a few core concepts. Items are represented as nodes, and the edges connecting them are partial correlations, that is, the association between any two items after the influence of every other item in the model has been statistically removed (Epskamp & Fried, 2018). An edge that survives this conditioning reflects genuine conditional dependence rather than shared correlation with a third variable. Nodes that are densely connected to one another tend to cluster into communities, which are analogous to factors in that they capture coherent groupings of items (Golino & Epskamp, 2017). The conceptual basis differs, though: communities emerge from the pattern of direct item-to-item relationships rather than from shared loadings on a latent dimension (Briganti et al., 2018;

Epskamp & Fried, 2018). A useful bridge between the two frameworks is that a true latent common cause would, in a network, manifest as a fully connected community in which every member node is associated with every other (Borsboom & Cramer, 2013; Epskamp & Fried, 2018). Recovering four communities broadly corresponding to Davis's (1980, 1983) subscales would therefore not contradict the four-factor structure; it would support it from a different angle. Beyond community structure, network analysis offers centrality indices that identify which nodes are most influential within the system. Strength centrality, the sum of a node's absolute edge weights, is generally the most stable of these indices across samples and estimation procedures, and it captures how embedded a given item is in the network as a whole (Epskamp, Borsboom & Fried, 2018).

The first application of this framework to the IRI was carried out by Briganti et al. (2018) in a sample of 1,973 French-speaking Belgian university students (mean age = 19.6 years; 57% women, 43% men). They estimated a Gaussian Graphical Model using the graphical LASSO with an extended Bayesian information criterion (EBICglasso) applied to Spearman correlations, and they used both spinglass and walktrap algorithms for community detection (Briganti et al., 2018). The spinglass algorithm recovered four communities aligned with Davis's subscales; walktrap suggested five, splitting off a small additional cluster of items. Strength centrality analysis identified an Empathic Concern item, specifically item 14, as the most central nodes in the network, although its centrality was not statistically distinguishable from a cluster of other highly connected items (Briganti et al., 2018). Mean node predictability was 0.27, indicating that a given item's neighbours explained around 27% of its variance. The prominence of Empathic Concern in the network has a natural precedent in the factor-analytic literature. In Cliffordson's (2002) hierarchical confirmatory factor analysis, the general empathy factor was empirically almost identical to Empathic Concern, with a standardised loading approaching 1.0. Perspective Taking and Fantasy emerged as weaker specifics, and Personal Distress as the least integrated subscale. If Empathic Concern anchors the hierarchical latent structure, we would expect EC items to be most densely connected in a network representation, and Briganti et al.'s (2018) findings are consistent with precisely that expectation.

The present study replicates this analysis in a new context. Where Briganti et al. (2018) drew on a French-speaking Belgian sample responding to a French adaptation of the IRI, the present study draws on an English-language US university sample using Davis's (1983) original English instrument, collected by Han (2024). The populations are comparable: undergraduate students with a mean age near 20, making it possible to assess whether the four-community structure and the centrality of Empathic Concern generalise across linguistic and cultural contexts. Establishing replication under these conditions raises the further question of whether the structure holds across subgroups, particularly between men and women.

Gender Differences in Empathy

Among the most frequently reported findings in this literature is a gender difference: on self-report measures, women tend to score higher than men, and the IRI has been central to documenting it. Across many studies, women report higher Fantasy, Empathic Concern and Personal Distress, while Perspective Taking tends to show little or no difference (Braun et al., 2015; Davis, 1980; Elkin et al., 2021; Grevenstein, 2020; Lucas-Molina et al., 2017; Mestre et al., 2009; Pang et al., 2023). This pattern does not appear to be an artefact of any single instrument or culture: it recurs in Spanish (Lucas-Molina et al., 2017; Mestre et al., 2009), German (Grevenstein, 2020), French (Braun et al., 2015) and Chinese (Pang et al., 2023) adaptations. Whether such differences reflect genuine latent variation rather than measurement bias has been tested directly. Two invariance studies both found that the IRI measures the same constructs in men and women, which suggests the score differences reflect real differences rather than measurement artefacts (Grevenstein, 2020; Lucas-Molina et al., 2017). Braun et al. (2015), by contrast, found only weak invariance, and noted that the factor inter-correlations themselves differed by gender, with the PT-EC and FS-EC associations stronger in women. This finding is notable because it suggests that gender may shape not only how much empathy people report but how its components are organised.

The affective and cognitive sides of the IRI do not behave alike here. Drawing on more than 101,000 respondents across 24 countries, Romero et al. (2026) measured only two of the four components, Perspective Taking and Empathic Concern, and found women scoring higher on both. The Empathic Concern gap tended to be larger and more stable across nations than the Perspective Taking gap, which varied more from one country to the next, a pattern they read as evidence that gender differences in affective empathy are the more universal of the two. Findings on cognitive empathy are inconsistent. Using a behavioural task rather than self-report, McDonald and Kanske (2023) found that women showed more empathy (affective sharing) and more compassion (the analogue of Empathic Concern) than men, but no gender differences in theory of mind (the analogue of cognitive Perspective Taking).

Why the gap exists is contested, the evidence supports two competing accounts. One account emphasises socialisation and gender roles: the self-report difference is, at least in part, conformity to gender stereotypes and social desirability, and it tends to diminish when the measurement of empathy is less transparent to participants. Several findings converge on this. The female advantage is large on transparent self-report scales but shrinks or disappears under less reactive measures. It is close to zero for physiological responses (Eisenberg & Lennon, 1983), absent in EEG responses to others' pain (Pang et al., 2023), and absent in an experimental empathy-for-pain task, even when the same participants show the usual gap on the IRI (Baez et al., 2017). Experimental manipulations support this account, with an empathy-inducing prime raising men's Empathic Concern to the level of women's (Pang et al., 2023), and Rajasekhar et al. (2025) reported that the difference, while real, was entangled with social desirability and gender-role identity. A competing account locates the difference in biology. Christov-Moore et al. (2014) argued that the female advantage is strong and consistent for affective empathy but weaker and mixed for cognitive empathy, pointing to phylogenetic and ontogenetic roots such as greater reactive crying in female neonates and the primary-caretaker hypothesis. Wu et al. (2022) found women higher specifically on Personal Distress, with that difference linked to the left anterior insula.

The two views do not need to be reconciled to identify their point of agreement: both locate the clearest female advantage in the affective components, Empathic Concern and Personal Distress (Christov-Moore et al., 2014; Mestre et al., 2009). Fantasy, though conventionally grouped with the cognitive scales, also tends to be higher in women, whereas Perspective Taking shows the smallest and most variable difference.

Gender and the Structure of Empathy

What almost all of this work has in common is a focus on mean levels. Whether the structure of empathy, the way its components interact, differs by gender has received far less attention; Briganti et al. (2018) flagged exactly this as a limitation of their own analysis. Network-comparison methods now make this question tractable. Studies across other domains have used them to detect gender differences in structure and global connectivity, including depression (Lee & Hu, 2022), eating disorders (Perko et al., 2019) and general mental health (Fan et al., 2025). The appropriate tool is the Network Comparison Test (NCT), a permutation procedure that compares two networks on overall structure, individual edges and global strength (van Borkulo et al., 2023).

The Present Study

The literature thus establishes robust mean-level differences but provides little evidence on whether the architecture of empathy itself differs by gender. That gap is what the present study addresses, by asking whether the network structure of empathy, the way the four IRI components interact, is the same for women and for men. We test three hypotheses.

H1 (Replication). The overall empathy network will replicate the structure reported by Briganti et al. (2018) in this independent dataset (Han, 2024), recovering four distinct communities corresponding to Davis's subscales, with Empathic Concern items showing the highest centrality (Briganti et al., 2018; Cliffordson, 2002).

H2 (Centrality by Gender). Personal Distress, Empathic Concern and Fantasy items will show higher centrality in women's networks than in men's, whereas Perspective Taking centrality will not differ significantly by gender (Elkin et al., 2021; Grevenstein, 2020; Lucas-Molina et al., 2017; Pang et al., 2023; Paulus, 2009; Wu et al., 2022).

H3 (Global Strength by Gender). Women's empathy network will show greater global connectivity than men's, reflecting stronger and more integrated affective empathic responses (Fan et al., 2025; Lee & Hu, 2022; Perko et al., 2019).

Method

Data source

This study is a secondary analysis of an existing dataset collected by Han (2024) and made openly available through the Open Science Framework (https://osf.io/qzftn/). Han's original study examined the network of moral-functioning components, and for the empathy measure reported factor scores for only two of the four IRI subscales, Empathic Concern and Perspective Taking. The archived data, however, contain participants' responses to all 28 IRI items. We analysed these item-level responses, allowing the empathy network to be estimated at the item level rather than from aggregated subscale scores.

Participants

The analysed sample comprised respondents with complete data on all 28 IRI items and a recorded gender, which yielded 1,468 students attending a public university in the southern

United States. The great majority were women (85.49%; *n* = 1,255), the rest were men (*n* = 213), and the mean age was 21.92 years (SD = 5.99). The replication analysis (H1) used the full sample. For the gender comparison (H2 and H3) the sample was divided into women and men.

Procedure

Participants signed up through a departmental research subject pool and completed the questionnaires online in a Qualtrics survey, receiving course credit in return. All study procedures and the online informed-consent form were reviewed and approved by the University of Alabama Institutional Review Board (protocol #18-12-1842). Besides the IRI, Han (2024) administered measures of moral reasoning, moral identity and purpose, along with a civic-engagement scale. These instruments fall outside the scope of the present study and are not considered further; only the IRI is described below.

Measure: Interpersonal Reactivity Index

Empathy was assessed with the Interpersonal Reactivity Index (IRI; Davis, 1983), in its original English form. The IRI is a 28-item self-report questionnaire made up of four seven-item subscales. Perspective Taking (PT) and Fantasy (FS) are conventionally treated as cognitive facets, while Empathic Concern (EC) and Personal Distress (PD) are treated as affective facets. Respondents indicate how well each statement describes them on a five-point scale, from “does not describe me well” to “describes me very well”. In the archived dataset these responses are coded from 1 to 5, rather than the 0 to 4 coding used in Davis’s original scoring; because the survey was administered in Qualtrics, which numbers its answer options from 1, the instrument itself is unchanged. Higher scores reflect a stronger standing on the tendency that each subscale captures.

Data

All data preparation was carried out in a single script that produces one analysis-ready file, so that the replication (H1) and the gender-comparison (H2, H3) analyses were run on identical data and the pipeline remains fully reproducible. From the eight data-collection waves archived by Han (2024) we retained the participants with complete responses on every IRI item, giving a sample size of 1,468. The 28 items were kept as individual nodes rather than aggregated into subscale totals: they followed Davis's original ordering, were mapped to nodes 1-28, and were each labelled by their a-priori subscale. Following Briganti et al.'s (2018) pipeline, the reverse-keyed items (3, 4, 7, 12, 13, 14, 15, 18, 19) were left in their raw direction rather than reverse-scored, so the estimated network legitimately contains negative edges. This feature informs several analytic decisions described below.

Analytic Approach

All analyses were conducted in R using the qgraph (Epskamp et al., 2012), bootnet (Epskamp, Borsboom & Fried, 2018), igraph (Csárdi & Nepusz, 2006), mgm (Haslbeck & Waldorp, 2020), and NetworkComparisonTest (van Borkulo et al., 2019) packages. A fixed random seed was set so that every step with a stochastic component (bootstrapping, permutation testing, subsampling) is reproducible, and the network-estimation settings were held identical across all analyses so that any observed between-group differences cannot be attributed to differences in estimation procedure.

Part 1: Replicating the Empathy Network (H1)

We estimated a regularised partial-correlation network (Gaussian graphical model) using the graphical LASSO with extended Bayesian information criterion (EBIC) model selection (EBICglasso; Epskamp & Fried, 2018; Foygel & Drton, 2010; Friedman et al., 2008), computed on a Spearman correlation matrix with pairwise-complete observations. The LASSO penalty shrinks small partial correlations to exactly zero, returning a sparse and interpretable network (Epskamp & Fried, 2018). We quantified node predictability, the proportion of each node's variance explained by its neighbouring nodes, using the mgm package (Haslbeck & Waldorp, 2020), providing an absolute measure of how strongly each item is determined by the rest of the network.

To address the replication claim about the network's community structure, we applied two community-detection algorithms to the absolute-valued edge weights: the spinglass algorithm, which identifies communities by assigning strongly interconnected nodes to the same cluster (Reichardt & Bornholdt, 2006), and the walktrap algorithm, which infers communities from patterns of short random walks through the network (Pons & Latapy, 2006). We then examined whether the recovered communities align with Davis's four theoretical subscales. We computed three centrality indices: strength, closeness, and betweenness. Strength is the sum of the absolute weights of all edges connecting a node to the rest of the network, and thus reflects how strongly a node is directly associated with the other items. Closeness is the inverse of the sum of the shortest path lengths between a node and all other nodes, indexing how readily a node can be reached from, and can reach, the rest of the network. Betweenness counts the number of shortest paths between other pairs of nodes that pass through the focal node, capturing the extent to which it lies on the indirect connections between items (Opsahl, Agneessens, & Skvoretz, 2010). We treated strength as the primary index, as closeness and betweenness are typically less stable than strength in psychometric networks (Epskamp & Fried, 2018; Epskamp, Borsboom & Fried, 2018). We then assessed the stability of the centrality and edge-weight estimates with a case-dropping bootstrap implemented in bootnet, summarised with the correlation-stability (CS) coefficient, which, as recommended by Epskamp, Borsboom and Fried (2018), should not fall below 0.25 and preferably exceed 0.50 for centrality findings to be interpreted. This analysis corresponds to the supplementary stability figures of Briganti et al. (2018).

Part 2: Gender Differences in the Empathy Network (H2 and H3)

To test for gender differences, we estimated separate networks for women and men using estimation settings identical to those in Part 1, and compared them with the Network Comparison Test (NCT), a permutation-based procedure for testing the invariance of network structure, edge strength, and global strength across two independent, cross-sectional samples (van Borkulo et al., 2019, 2023). Following van Borkulo et al. (2023), we used the test of global-strength invariance (Opsahl et al., 2010) to evaluate H3. To evaluate H2, we used NCT's per-node centrality comparison. Unlike the structure, edge-strength, and global-strength tests, which were validated in the simulations of van Borkulo et al. (2023), this node-level extension was not evaluated there. We therefore additionally supported its interpretability through the per-group stability and equal-n analyses described below. We ran 5,000 permutations to obtain stable p-values. The group order was (women, men), so a positive difference indicates Women > Men, consistent with both directional hypotheses. The directional, family-level test of H2 is conducted on strength centrality. Given that the node-level hypotheses are directional and specified a priori for three item families (Empathic Concern, Personal Distress, and Fantasy predicted to be more central in women, Perspective Taking predicted not to differ). We therefore applied the Bonferroni-Holm correction across node comparisons (Holm, 1979; van Borkulo et al., 2023) and reported each family explicitly rather than collapsing the results.

Sensitivity and Robustness Analyses

The marked imbalance between groups (1,255 women vs. 213 men) is the central methodological concern for the gender comparison and is treated as part of the primary analysis rather than as a secondary consideration. Because EBICglasso regularisation is sample-size dependent, smaller samples are shrunk more aggressively. As a result, they tend to yield sparser networks with lower global strength and centrality, even when these differences are solely attributable to sample size. Van Borkulo et al. (2023) highlight this issue in their male-versus-female application, noting that apparent differences in density and strength “could be due to the larger sample size of the female group”. We therefore conducted an equal-n subsampling sensitivity analysis in which the women were repeatedly down-sampled to the men's n, the network re-estimated under identical settings, and global strength and node strengths recorded across replications. We led the reporting of H3 with this equal-n result: if the difference attenuates once the groups are equated in size, we interpret the raw difference as substantially attributable to sample size.

Two further safeguards supported the node-level claims. First, H2's centrality claims are only interpretable when centrality is stably estimated within each group. We therefore ran case-dropping bootstraps separately for the women's and men's networks (Epskamp, Borsboom & Fried, 2018). The men's network is the limiting case: if its strength CS-coefficient falls below 0.25, we report the corresponding centrality differences only qualitatively. Second, as a complementary check that uses the full sample and avoids regularising the small male subsample in isolation, we estimated a moderated network model with gender as a binary moderator (Haslbeck et al., 2021), implemented in mgm (Haslbeck & Waldorp, 2020), which estimated gender-by-edge interactions directly.

Reporting

We reported H1 through the estimated network, the centrality plot, the comparison of detected communities against the four subscales, and the bootstrapped stability analyses. We reported H2 through the per-node strength differences from the NCT, gated by the per-group CS-coefficients and cross-checked at equal *n*, presented by item family. We reported H3 through the NCT global-strength test alongside the equal-*n* subsampling result, leading with the latter. The omnibus network-structure invariance test was reported together with the main gender-comparison results. Throughout, the gender imbalance was acknowledged as a limitation, and the H2 and H3 findings were framed with appropriate caution regarding the statistical power of the male network.

Results

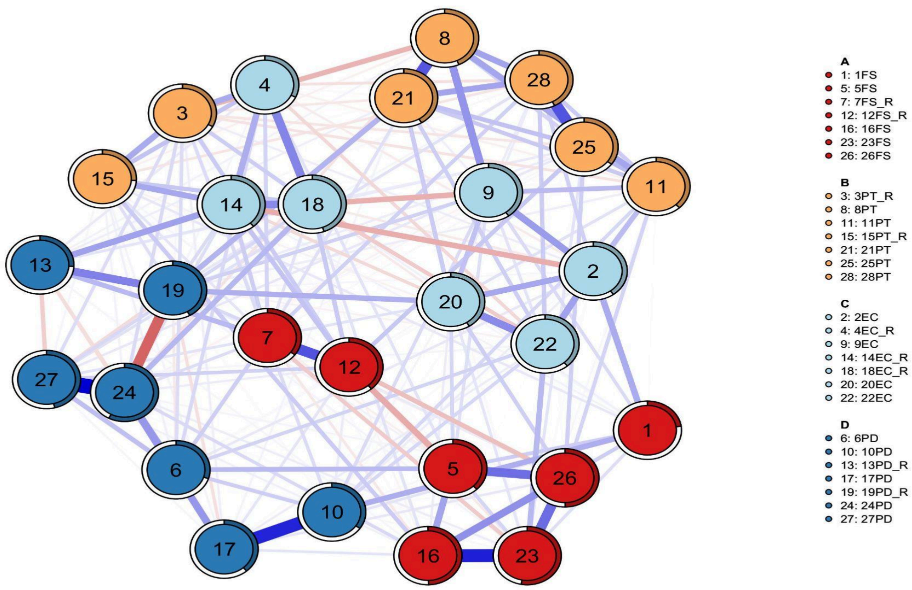
Network Structure in the Full Sample

Network Estimation

The empathy network was estimated using a Gaussian Graphical Model (GGM) with EBICglasso regularisation based on Spearman correlations (Friedman et al., 2008). A low regularisation parameter ($\lambda < 0.10 \times \lambda\_max$), yielded a dense network in which 378 of the possible edges were retained. The mean edge weight was $M = 0.027$. Nodes were labelled according to the original IRI item numbering. The suffix "_R" indicates reverse-worded items. The strongest edges, as observed in Figure 1, were between item 24 and 27 (Personal Distress; PD), item 16 and 23 (Fantasy; FS), item 25 and 28 (Perspective Taking; PT), item 8 and 21 (Perspective Taking; PT), and item 7 and 12 (Fantasy; FS_R; reversed items).

Figure 1

Network analysis results in the full sample



Note. This figure shows the network in the full sample; the subfigures display the separate network components or comparisons. Nodes represent the 28 IRI items. Blue edges indicate positive associations and red edges indicate negative associations. Edge thickness reflects association strength. Node colours indicate the detected communities.

Node Predictability

Mean node predictability was $R^2 = 0.39$ ($SD = 0.08$), indicating that on average 39% of the variance in each item was explained by the remaining items in the network. The highest predictability values were observed for item 24 (PD; $R^2 = 0.58$), item 23 (FS; $R^2 = 0.53$), item 26 (FS; $R^2 = 0.50$), and item 16 (FS; $R^2 = 0.50$). The lowest predictability values were observed for item 1 (FS; $R^2 = 0.23$) and item 13 (PD_R; $R^2 = 0.26$).

Community Detection

Community structure was examined using the spinglass algorithm run 100 times with different random seeds. On average, 4.58 communities were detected (*Mdn* = 4.5; range =

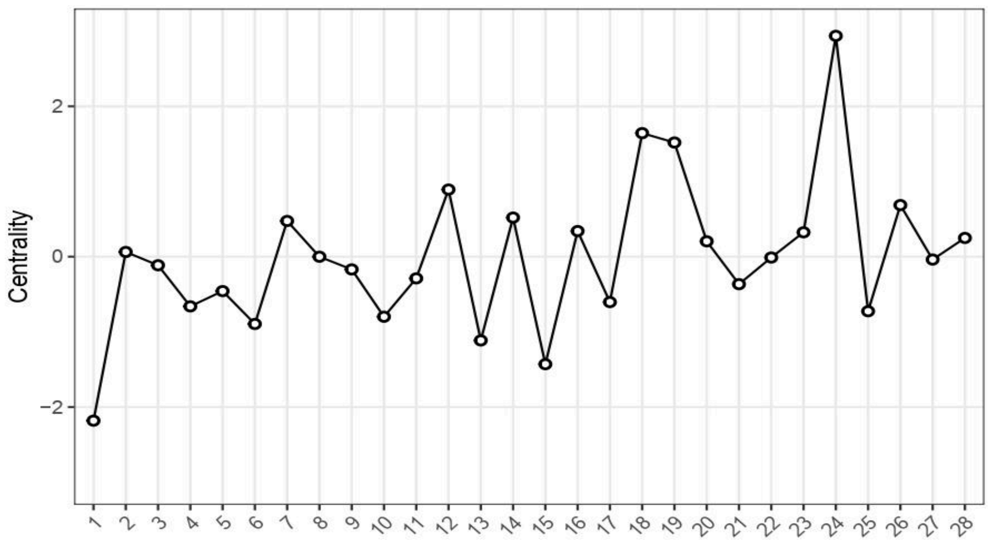
4-6). In the primary solution (seed = 1), five communities were identified. The walktrap algorithm also identified a fifth community consisting of items 6, 10, and 17, all belonging to the Personal Distress subscale.

Centrality Indices

Node centrality as illustrated in Figure 2 was assessed using strength, closeness, and betweenness. The highest strength centrality values were observed for item 24 (PD; 1.56), item 18 (EC_R; 1.35), and item 19 (PD_R; 1.33), followed by item 12 (FS_R; 1.23), item 26 (FS; 1.20), and item 14 (EC_R; 1.17). The lowest strength centrality values were observed for item 1 (FS; 0.74), item 15 (PT_R; 0.86), and item 13 (PD_R; 0.91). Spearman correlations among centrality indices were moderately intercorrelated: strength-betweenness $r = .55$, strength-closeness $r = .57$, and closeness-betweenness $r = .57$.

Figure 2

Node Strength Centrality in the Empathy Network



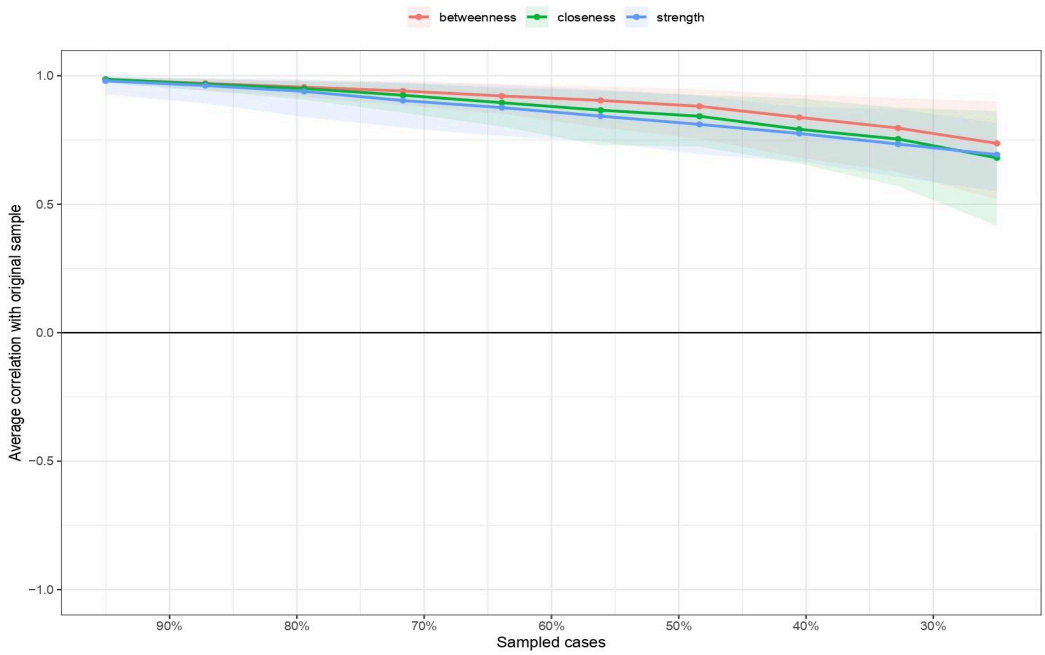
Note. Higher values indicate stronger connections of a node with the rest of the network. Item 24 (PD) showed the highest strength centrality, followed by items 18 (EC_R) and 19 (PD_R).

Stability of Centrality Indices

Stability was assessed using a case-dropping bootstrap procedure (2,000 iterations; Epskamp, Borsboom & Fried, 2018). The correlation stability (CS) coefficients were 0.52 for strength, 0.52 for closeness, 0.59 for betweenness, and 0.75 for edge weights. All CS coefficients in the full sample exceeded the recommended minimum threshold of 0.25 and the preferred threshold of 0.50, indicating good stability of the network estimates as illustrated in Figure 3.

Figure 3

Case-Dropping Bootstrap Stability of Centrality Indices



Note. Case-dropping bootstrap results showing the average correlation between centrality estimates obtained from subsamples and the original sample. Shaded areas represent 95% confidence intervals.

Node-Specific Centrality Differences by Gender

Stability by Gender

Separate networks were estimated for women (n = 1,255) and men (n = 213). In the female sample, the network showed acceptable stability, with CS-coefficients of 0.75 for strength, 0.52 for closeness, and 0.59 for betweenness. In contrast, stability in the male sample was poor, with CS = 0.13 for strength and CS = 0.00 for both closeness and betweenness. These values fall below the 0.25 minimum recommended by Epskamp, Borsboom & Fried (2018), so centrality estimates for men are not interpretable

Node-Specific Centrality Differences

Descriptively, all 28 nodes showed higher strength centrality in women compared to men. The largest differences (women minus men) were observed for item 19 (PD_R; Δ = 1.06), item 14 (EC_R; Δ = 0.89), item 3 (PT_R; Δ = 0.83), item 24 (PD; Δ = 0.83), and item 13 (PD_R; Δ = 0.77). However, after Bonferroni-Holm correction for multiple comparisons (Holm, 1979; van Borkulo et al., 2023), none of these differences reached statistical significance (all adjusted p = 1.00). Consequently, no subscale showed any significant gender differences in node strength centrality.

Global Network Strength by Gender

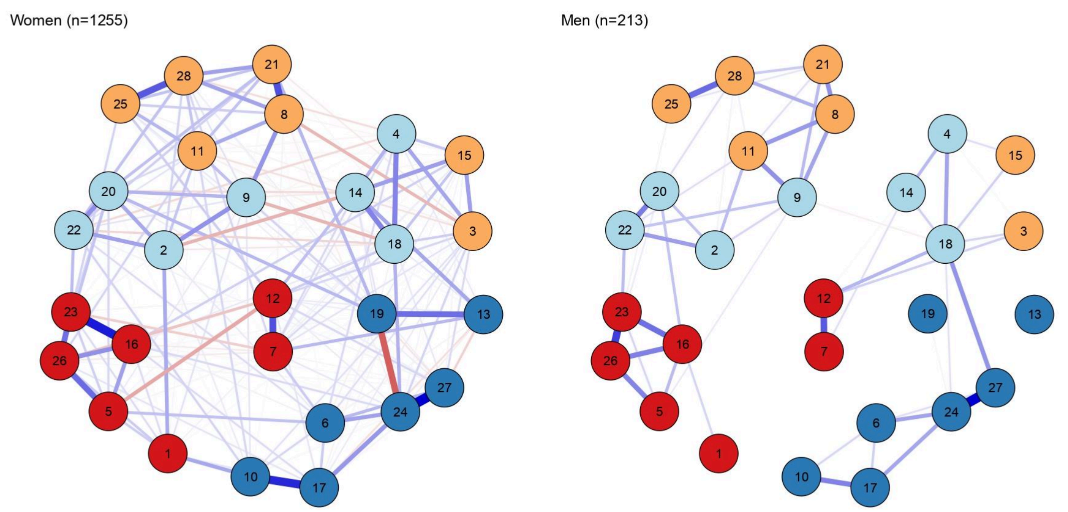
Sensitivity Analysis with Equal Sample Sizes

To address potential sample size imbalance, 1,000 bootstrap subsamples of n = 213 were drawn from the female group. The median global strength in the female subsamples was 8.93 (95% range: 4.06-11.60), compared to 4.91 in the male sample. As the male value falls within the 95% range of the equal-n female subsamples, the global-strength difference disappeared once group sizes were equated. At the node level, the mean strength of the female subsamples exceeded the male values across all 28 nodes. The largest node-level differences were item 14 (EC_R; Δ = 0.46), item 20 (EC; Δ = 0.39), item 19 (PD_R; Δ = 0.52), item 24 (PD; Δ = 0.38), item 23 (FS; Δ = 0.25), and item 26 (FS; Δ = 0.25).

The Network Comparison Test (NCT; van Borkulo et al., 2019) was conducted using 5,000 permutations to assess gender differences in network structure. The comparison of global strength did not reveal a significant difference. Although women showed a higher global strength (13.76) than men (4.91), resulting in an observed difference of 8.85 (Figure 4), this difference did not reach statistical significance according to the NCT (p = .085).

Figure 4

Gender Differences in Empathy Network Structure and Global Strength



Note. Separate networks are shown for women (n = 1,255) and men (n = 213). The histogram displays the permutation distribution of global strength differences obtained from the Network Comparison Test.

Network Structure Invariance (Omnibus)

The omnibus network-structure invariance test yielded a maximum edge-difference statistic of $M = 0.257$, which reached statistical significance ($p = .047$). This indicates that at least some connections differed between the two networks.

Moderated Network Model

A moderated network model was estimated on the full sample (N = 1,468), with gender included as a binary moderator. Gender showed direct associations with item 2 (EC; *I often have tender, concerned feelings for people less fortunate than me*) and item 17 (PD; *Being in a tense emotional situation scares me*). Potential moderation effects were examined by estimating gender-by-edge interaction parameters for all network connections. No significant gender-by-edge interactions were detected. No edge-by-gender interaction parameters were nonzero, indicating no moderation of item-item associations by gender.

Discussion

The present study investigated the structure of the Interpersonal Reactivity Index (IRI) using network analysis and examined whether the organisation of empathy differs between men and women. Overall, the findings partially supported the hypothesised structure of the IRI, replicating a coherent and interpretable empathy network in a large university sample. However, evidence for gender differences in network structure and connectivity was weak and should be interpreted with caution due to methodological constraints.

Replication of the IRI Network Structure

In line with previous research (Briganti et al., 2018; Lucas-Molina et al., 2017; Raimondi et al., 2023), the present study broadly replicated a four-community structure corresponding to the original IRI subscales, providing partial support for H1. This supports the idea that the IRI reflects a meaningful organisation of interrelated components rather than independent latent dimensions. However, some deviations were observed, including the emergence of a fifth community consisting of Personal Distress items. Similar deviations from the classical four-factor structure of the IRI have been reported in previous studies, particularly involving instability of the Personal Distress subscale and variability in its clustering across samples (Briganti et al., 2018; Lucas-Molina et al., 2017; Raimondi et al., 2023). This may indicate that Personal Distress represents a more heterogeneous construct or that its internal structure is particularly sensitive to sample or estimation choices.

Briganti et al. (2018) found Empathic Concern items among the most central nodes. in IRI-based network analyses. In the present study, by contrast, affective items (EC and PD) were most central, with a Personal Distress item the most central of all. This does not support the part of H1 predicting that Empathic Concern items would show the highest centrality. The broader pattern suggests nonetheless that affective components of empathy contribute prominently to the overall system. In network terms, strong communities may reflect underlying latent constructs, which manifest as densely interconnected clusters of items (Epskamp, Borsboom & Fried, 2018; Kruis & Maris, 2016).

Gender Differences in Empathy Network

A secondary aim of this study was to examine whether the structure and connectivity of empathy differs between men and women (H2, H3). Descriptively, women showed higher node-level centrality across all items. Because Perspective Taking items were also higher, this pattern does not match our H2 prediction that Perspective Taking would not differ by gender. These effects were not statistically significant after correction; moreover, the uniform elevation across all nodes is the pattern expected from the larger women's sample and the sample-size dependence of EBICglasso regularisation, rather than evidence of a substantive global difference. Women also showed descriptively higher global network strength, but this difference fell within sampling variation once group sizes were matched, so it does not provide confirmatory support for H3.

In contrast, the omnibus Network Comparison Test was significant, suggesting that the overall pattern of associations among empathy items may differ between men and women, even in the absence of differences in global connectivity. However, the moderated network model identified no significant gender moderation effects on individual network edges. Although gender showed direct associations with one Empathic Concern item and one Personal Distress item, no specific associations between empathy items differed significantly as a function of gender. Taken together, these findings suggest that any gender-related differences in network structure are not attributable to a small number of distinct connections. Instead, previously reported gender differences in empathy may primarily reflect mean-level differences in empathy items rather than structural differences in how empathy components are interconnected (Baez et al., 2017; Eisenberg & Lennon, 1983; Pang et al., 2023).

Methodological Considerations in Gender Comparisons

Several methodological factors likely constrain the interpretation of these findings. First, the gender distribution was highly imbalanced (approximately 85% women vs. 15% men). Because EBICglasso regularisation is sensitive to sample size, differences in group size may influence network estimation, with smaller groups potentially yielding less stable and sparser network solutions (Epskamp & Fried, 2018; van Borkulo et al., 2023). This may partly account for observed differences in global connectivity and is likely to reduce statistical power and stability in the male subsample. To address this issue, a resampling procedure with 1,000 iterations matching the male sample size was conducted to assess the robustness of the observed network differences. When sampling sizes were equalised, the male global strength fell within the range of the down-sampled female subsamples, indicating that a substantial portion of the apparent difference was driven by sample size and regularisation rather than by a genuine structural difference.

Second, the male network showed poor stability, with very low correlation stability coefficients for centrality indices. This indicates that centrality estimates in this group are not reliable and should be interpreted with caution (Epskamp, Borsboom & Fried, 2018).

Third, while the Network Comparison Test (van Borkulo et al., 2023) is the recommended approach for group comparisons, power in such analyses is largely determined by the smaller group. Given the relatively small male subsample, null findings may reflect insufficient power rather than true equivalence between groups.

A further methodological issue concerns the construction of networks from correlational data. Simulation studies have demonstrated that standard network metrics including centrality and small-worldness can arise as artefacts of the thresholding process, even when the underlying correlations are essentially random (Papo et al., 2016; Zalesky et al., 2012). In other words, part of what may be observed in such networks can reflect artefacts of the estimation procedure rather than anything psychologically substantive. Although the regularised estimation approach used here (EBICglasso) is more sophisticated than simple thresholding, it is not immune to similar concerns: edges are retained or dropped based on statistical criteria that do not necessarily correspond to meaningful psychological relationships. A related concern is that much of the pairwise covariation observed in psychometric networks may reflect item redundancy and shared latent influences rather than direct interactions between psychological components (Waldorp & Marsman, 2021; Marinazzo et al., 2024). Consequently, strong edges and high centrality may partly reflect overlapping item content rather than the functional organisation of empathy as a system. This underscores the importance of treating centrality values as descriptive indicators rather than firm evidence of a node's psychological influence.

Finally, multiple methodological constraints further limit strong conclusions about group differences. Bootstrapped edge comparisons are not ideally suited for cross-group inference, and centrality confidence intervals are generally not recommended due to biased sampling distributions (Epskamp, Borsboom & Fried, 2018). Instead, stability coefficients provide a more appropriate metric, which in the present study indicated acceptable stability only in the female subsample. Taken together, these limitations suggest that the observed gender differences in network connectivity should be interpreted as descriptive trends rather than robust inferential findings.

Methodological Considerations of Network Analysis

More broadly, several characteristics of network analysis influence the interpretation of the present findings. First, sparsity in LASSO-regularised networks does not imply the absence of a relationship (Epskamp & Fried, 2018; van Borkulo et al., 2023). Rather, edge selection is influenced by regularisation criteria that prioritise specificity over sensitivity, meaning that true associations may be omitted (false negatives) (van Borkulo et al., 2023). Second, network and latent-variable models are not mutually exclusive frameworks. In fact, latent constructs may manifest as strongly interconnected communities within a network structure. The observed alignment between IRI subscales and detected communities therefore likely reflects this conceptual overlap rather than evidence favouring one model over the other. Third, the present study relies on cross-sectional data, which prevents any conclusions about directionality or temporal dynamics of empathic processes. As in previous IRI network studies, the observed edges represent statistical associations rather than causal pathways (Borsboom & Cramer, 2013; Epskamp & Fried, 2018). Additionally, the use of self-report measures introduces potential bias, including social desirability effects and shared method variance, which may inflate associations between items (Baez et al., 2017; Eisenberg & Lennon, 1983; Pang et al., 2023; Rajasekhar et al., 2025). Importantly, previous research suggests that such biases may be particularly relevant in empathy measurement (Eisenberg & Lennon, 1983; Pang et al., 2023). Self-reported empathy, especially on the Interpersonal

Reactivity Index, tends to yield larger gender differences than behavioural or physiological indices (Eisenberg & Lennon, 1983; Pang et al., 2023).

Finally, an important limitation concerns the decision to leave reverse-keyed items un-reverse-scored. Even though this was adopted to match Briganti et al.'s (2018) pipeline, it carries analytic costs. This may have contributed to the presence of negative edges in the network, potentially reflecting method effects associated with reverse-worded items and measurement artefacts rather than substantive associations (Epskamp & Fried, 2018; Paulus, 2009). In addition, given known psychometric complexity of the IRI, including cross-loadings and unstable item functioning across studies, network structure estimates may partially reflect measurement artefacts rather than purely substantive psychological processes (Grevenstein, 2020; Lucas-Molina et al., 2017; Raimondi et al., 2023). Therefore, all network interpretations should be considered exploratory and interpreted with caution.

Looking ahead, it will be important for researchers to consider how much of what we call "network structure" in psychometric data actually reflects substantive psychological organisation, and how much is an artefact of the methods we use. One useful step would be to validate network metrics against simulated data where the ground truth is known (as in Papo et al., 2016; Zalesky et al., 2012). Another would be to explicitly model the contribution of latent variables, following the framework proposed by Waldorp and Marsman (2021). Until such validation becomes routine, centrality and connectivity estimates in psychometric networks should be interpreted cautiously, especially where strong edges may reflect items with near-identical content.

Implications and Future Research

Despite these limitations, the present findings contribute to the emerging literature on network approaches by further supporting a largely stable four- to five-community structure broadly consistent with the multidimensional organisation of the Interpersonal Reactivity

Index (Davis, 1983; Grevenstein, 2020; Lucas-Molina et al., 2017; Raimondi et al., 2023). Future research should address the limitations observed here by using more balanced gender samples and longitudinal designs. Additionally, applying network approaches to clinical populations may be particularly informative, given evidence linking Personal Distress to psychological distress, depression, and neuroticism (Davis, 1983; Grevenstein, 2020; Paulus, 2009), and prior network studies linking empathy dimensions to clinical traits, such as schizotypy (Wang et al., 2020).

Finally, as highlighted in recent methodological discussions (Epskamp, Borsboom & Fried, 2018; van Borkulo et al., 2023), improving the statistical foundations of group comparisons in network analysis remains an important direction for future research, particularly when group sizes are unequal.

Conclusion

In sum, H1 was partially confirmed: a largely stable four-community structure broadly matching the IRI subscales emerged, alongside an additional Personal Distress community. H2 was not confirmed, as no statistically robust gender differences in node centrality were observed after correction for multiple comparisons. H3 was also not confirmed; although women showed descriptively higher global network strength, this difference was not statistically significant, and at equal *n* it was no longer clearly distinguishable. Given the underpowered male network, this is inconclusive rather than evidence of equivalence.

Overall, the findings provide only limited evidence for gender differences in the organisation of empathic processes. Methodological constraints, particularly sample imbalance and stability issues in the male subsample, suggest that conclusions regarding gender differences should remain tentative. Finally, it is worth noting that recent methodological work casts some doubt on whether centrality indices in psychometric networks can be straightforwardly interpreted as markers of psychological importance. Evidence that similar network properties emerge from thresholded random matrices (Papo et al., 2016; Zalesky et al., 2012), combined with findings that pairwise associations in these networks are largely attributable to latent factor overlap (Waldorp & Marsman, 2021), suggests that the present results are best understood as descriptive patterns in item covariation. They should not be taken as strong evidence for the causal role of any empathy component.

AI-Statement

Generative artificial intelligence (Claude, Anthropic) was used as a programming assistant during the preparation of the analysis scripts. Specifically, it was employed to help update archived R scripts to ensure compatibility with current software environments. This included resolving deprecated function calls, adapting package-specific syntax, and replacing file paths that referred to the original author's local machine. Claude was also used to assist in the implementation and debugging of code for the additional analyses described in this research project. All analytical decisions, statistical procedures, code modifications, and interpretations were determined, reviewed, and independently verified by the authors.

In addition, AI-assisted tools, including Julius AI, Paperguide, and Claude (Anthropic), were used during the preparation of this research paper to support the rephrasing, language editing, and academic refinement of the introduction, methods, results, and discussion sections. These tools functioned solely as linguistic and stylistic aids and did not contribute to the development of the study design, analyses, interpretations, or conclusions. The authors retain full responsibility for all intellectual content presented in this work.

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